Quadratic $(n, n+2, n)$ mass-action networks with $n+1$ positive nondegenerate equilibria

chemistry-open-problems · 18 October 2026 · verified; independent rediscovery of an earlier answer

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Abstract. Banaji and Feliu proved that a full-rank quadratic mass-action network with n species and $n+2$ distinct sources has at most $n+1$ positive nondegenerate equilibria, and wrote that they could not find a quadratic $(3,5,3)$ network with four, leaving open whether the bound is attained for $n \geq 3$. We give, for every $n \geq 1$, an explicit quadratic $(n, n+2, n)$ network whose sources are $0, 2X_1, X_1+X_2, \ldots, X_{n-1}+X_n, X_n$ and which has exactly $n+1$ positive nondegenerate equilibria for suitable rate constants. Its kernel can be prescribed freely, which reduces the count to the zeros of an explicit one-variable function; geometrically separated kernel ratios then force the maximal number of equilibria. For $n=1$ the network is $0 \to X,\ X \to 0,\ 2X \to 3X$, which shows that the bound is attained for $n=1$ as well, contrary to Remark 5.6 of the same paper. *The answer is not new* (Section 5): the equilibrium equations of these networks are, up to scaling, the extremal fewnomial chain of Phillipson and Rojas Theorem 1.6 [1], and a mass-action realization of that chain answering Remark 5.8 for every $n \geq 2$ was posted, unrefereed, on 4 October 2026 [2]. The present note is an apparently independent rediscovery, with a short self-contained proof.

The question

A network is *quadratic* if all its sources have molecularity at most two (products are unrestricted), and an (n, m, r) network has n species, m reactions and rank r [3]. Theorem 5.5 of [3] states that a dynamically nontrivial quadratic $(n, n+k, n)$ network with $n+2$ distinct sources which is not source deficient admits no more than $n+1$ positive nondegenerate equilibria. In Remark 5.8 the authors write (arXiv:2409.06877v5, p. 41):

> *"However, we were not able to find a quadratic $(3,5,3)$ network admitting four positive nondegenerate equilibria, and thus achieving the bound in Theorem 5.5. We leave open the question of whether the bound in Theorem 5.5 can be achieved by quadratic $(n, n+2, n)$ networks for $n \geq 3$."*

Theorem 1. For every $n \geq 1$ there is a quadratic $(n, n+2, n)$ mass-action network with $n+2$ distinct sources, dynamically nontrivial and not source deficient, which for suitable rate constants has exactly $n+1$ positive equilibria, all of them nondegenerate. Hence the bound of Theorem 5.5 [3] is attained for every $n \geq 1$ (with $k=2$).

Remark 2 (on Remark 5.6 of [3])**.** Remark 5.6 of [3] states that for $n=1$ the bound is not optimal, because "by [5, Lemma 20], quadratic rank-one networks do not admit multiple positive nondegenerate equilibria". Lemma 20 of the cited paper [4] says the opposite: a quadratic rank-one network admits multiple positive nondegenerate

equilibria if and only if it has at most two nontrivial species and, after removal of trivial species, contains three reactions forming a network simply equivalent to one of four small networks, the first being $0 \to X,\ X \to 0,\ 2X \to 3X$. The statement quoted in Remark 5.6 holds for *bimolecular* rank-one networks Corollary 19 [4]. Indeed, with rate constants 1, 3, 1 for $0 \to X,\ X \to 0,\ 2X \to 3X$ respectively, $\dot{x} = 1 - 3x + x^2$ has the two positive simple roots $(3 \pm \sqrt{5})/2$. This network is $\mathcal{N}_1$ below. (The same discrepancy is pointed out in the prior-art notes accompanying [2].)

1 The networks

Fix $n \geq 1$, species $X_1, \dots, X_n$, and positive integers $\alpha_2, \dots, \alpha_{n+1}$ and $\beta_2, \dots, \beta_{n+1}$. Write $\alpha = (1, 0, \alpha_2, \dots, \alpha_{n+1})$ and $\beta = (0, 1, \beta_2, \dots, \beta_{n+1})$. The network $\mathcal{N}_n(\alpha, \beta)$ has the $n+2$ reactions

$$R_0:\ 0 \longrightarrow \textstyle\sum_{i=1}^{n} \alpha_{i+1} X_i, \qquad R_1:\ 2X_1 \longrightarrow 2X_1 + \sum_{i=1}^{n} \beta_{i+1} X_i,$$
$$R_{i+1}:\ X_i + X_{i+1} \longrightarrow X_{i+1} \quad (1 \leq i \leq n-1), \qquad R_{n+1}:\ X_n \longrightarrow 0.$$

Its sources $\sigma_0 = 0$, $\sigma_1 = 2X_1$, $\sigma_{i+1} = X_i + X_{i+1}$, $\sigma_{n+1} = X_n$ are distinct and quadratic, and the products are nonnegative. Let $\Gamma \in \mathbb{Z}^{n \times (n+2)}$ be the stoichiometric matrix (columns indexed $0, \dots, n+1$) and $S \in \mathbb{Z}^{(n+2) \times n}$ the source matrix. The i-th row of Γ is $\alpha_{i+1} e_0 + \beta_{i+1} e_1 - e_{i+1}$.

Lemma 3. $\mathcal{N}_n(\alpha, \beta)$ is a full-rank $(n, n+2, n)$ network with $\ker \Gamma = \operatorname{span}(\alpha, \beta)$ and $\ker \Gamma \cap \mathbb{R}^{n+2}_{>0} = \{s\alpha + t\beta : s, t > 0\}$; in particular it is dynamically nontrivial. Moreover $\operatorname{rank}[S \,|\, \mathbf{1}] = n+1$, and the left kernel of $[S \,|\, \mathbf{1}]$ is spanned by $w = (w_0, w_1, \dots, w_{n+1})$ with

$$w_1 = 1, \qquad w_j = 2(-1)^{j-1}\ (2 \leq j \leq n+1), \qquad w_0 = -\textstyle\sum_{j \geq 1} w_j.$$

Proof. Row i applied to α gives $\alpha_{i+1} \cdot 1 + \beta_{i+1} \cdot 0 - \alpha_{i+1} = 0$, and to β gives $0 + \beta_{i+1} - \beta_{i+1} = 0$. The entries -1 sit in distinct columns $2, \dots, n+1$, so $\operatorname{rank} \Gamma = n$ and $\ker \Gamma$ is the two-dimensional space $\operatorname{span}(\alpha, \beta)$. Since $\alpha_1 = \beta_0 = 0$ and all other entries are positive, $s\alpha + t\beta > 0$ iff $s, t > 0$. The sources $X_n, X_{n-1} + X_n, \dots, X_1 + X_2$ span $\mathbb{R}^n$ and $\sigma_0 = 0$, so $\operatorname{rank}[S|\mathbf{1}] = n+1$. Finally $\sum_j w_j \sigma_j = 0$: the coefficient of X_1 is $2w_1 + w_2 = 2 - 2 = 0$, and that of X_i $(2 \leq i \leq n)$ is $w_i + w_{i+1} = 0$; and $\sum_j w_j = 0$ by the choice of w_0. □

2 Equilibria as roots of one function

Put $v(u) = \alpha + u\beta$ for $u > 0$, so $v_0 = 1$, $v_1 = u$, $v_j = \alpha_j + u\beta_j$ $(j \geq 2)$, and

$$f(u) = \sum_{j=0}^{n+1} w_j \log v_j(u) = \log u + \sum_{j=2}^{n+1} 2(-1)^{j-1} \log(\alpha_j + \beta_j u).$$

Lemma 4. Let $\kappa \in \mathbb{R}^{n+2}_{>0}$ and $c = \sum_j w_j \log \kappa_j$. The positive equilibria of $\mathcal{N}_n(\alpha, \beta)$ with rate constants κ are in bijection with the solutions $u > 0$ of $f(u) = c$.

Proof. $x > 0$ is an equilibrium iff the rate vector $\rho_j = \kappa_j x^{\sigma_j}$ lies in $\ker \Gamma \cap \mathbb{R}^{n+2}_{>0}$, i.e. $\rho = s\, v(u)$ for unique $s, u > 0$ (Lemma 3). Taking logarithms, this is the linear system

$\sigma_j \cdot \log x - \log s = \log v_j(u) - \log \kappa_j$ $(0 \le j \le n+1)$ in the $n+1$ unknowns $(\log x, \log s)$, whose coefficient matrix $[S| - \mathbf{1}]$ has rank $n+1$ and left kernel $\operatorname{span}(w)$. It is solvable iff $\sum_j w_j(\log v_j(u) - \log \kappa_j) = 0$, i.e. $f(u) = c$, and then its solution is unique. □

Lemma 5. The equilibrium corresponding to u is nondegenerate (its Jacobian is nonsingular) iff $f'(u) \neq 0$.

Proof. The Jacobian is $J = \Gamma D_\rho S D_{1/x}$ with $\rho = s\,v(u)$. As S is injective, J is singular iff $\operatorname{im} S \cap D_\rho^{-1} \ker \Gamma \neq \{0\}$. The left kernel of S is two-dimensional and contains w; since $\mathbf{1} \notin \operatorname{im} S$, it also contains some w' with $w' \cdot \mathbf{1} = 1$, and $\operatorname{im} S = \{q : w \cdot q = w' \cdot q = 0\}$. For $a\alpha + b\beta = a\,v(u) + (b - au)\beta$ we get $q = D_\rho^{-1}(a\alpha + b\beta) = s^{-1}(a\mathbf{1} + (b - au)(\beta_j/v_j)_j)$, hence $w \cdot q = s^{-1}(b - au)f'(u)$ and $w' \cdot q = s^{-1}(a + (b - au)\sum_j w'_j\beta_j/v_j)$. If $f'(u) \neq 0$, both vanish only for $a = b = 0$; if $f'(u) = 0$, $b - au = 1$, $a = -\sum_j w'_j\beta_j/v_j$ gives a nonzero $q \in \operatorname{im} S$. □

Lemma 6. For any c, $f - c$ has at most $n+1$ zeros in $(0, \infty)$, and if it has $n+1$ they are all simple.

Proof. $f'(u) = 1/u + \sum_{j=2}^{n+1} w_j\beta_j/(\alpha_j + \beta_j u)$. Multiplying by $u\prod_{j\ge 2}(\alpha_j + \beta_j u)$ gives a polynomial of degree at most n whose coefficient of u^n is $\prod_j \beta_j \cdot (1 + \sum_{j\ge 2} w_j) = \pm\prod_j \beta_j \neq 0$. So f' has at most n zeros, and by Rolle's theorem $f - c$ has at most $n+1$ zeros. If it has $n+1$, Rolle's theorem gives n zeros of f' strictly between them, which are then all the zeros of f'; so none of the zeros of $f - c$ is a zero of f'. □

3 Geometrically separated kernels

Fix $R \ge 64$ and take $\alpha_j = R^{j-1}$, $\beta_j = 1$ for $2 \le j \le n+1$; write $L = \log R$. Then $f(u) = \log u + \sum_{m=1}^{n} 2(-1)^m \log(R^m + u)$.

Proposition 7. Let $M_1 = f$'s "main value" at $u = R$, namely $M_1 = L(1 + 2\sum_{m=1}^{n}(-1)^m \max(m, 1))$, and $c = M_1 - L/2$. Then $f(R^i) - c > 0$ for odd i and $f(R^i) - c < 0$ for even i $(1 \le i \le n)$. Moreover $f(u) \to -\infty$ as $u \to 0^+$, and $f(u) \to -\infty$ (n odd) or $+\infty$ (n even) as $u \to \infty$. Consequently $f - c$ has exactly $n+1$ zeros in $(0, \infty)$, all simple.

Proof. For $1 \le i \le n$, $\log(R^m + R^i) = \max(m, i)L + \log\left(1 + R^{-|m-i|}\right)$. Hence $f(R^i) = M_i + 2(-1)^i \log 2 + \varepsilon_i$ with $M_i = L(i + 2\sum_{m=1}^{n}(-1)^m \max(m, i))$ and $|\varepsilon_i| \le \sum_{d\ge 1} 2 \cdot 2\log\left(1 + R^{-d}\right) \le 4/(R - 1)$ (each $d = |m - i| \ge 1$ occurs at most twice). A direct computation gives $M_{i+1} - M_i = L(1 + 2\sum_{m=1}^{i}(-1)^m)$, which is $-L$ for odd i and $+L$ for even i; so $M_i = M_1$ for odd i and $M_i = M_1 - L$ for even i. Therefore

$$f(R^i) - c = \begin{cases} L/2 - 2\log 2 + \varepsilon_i, & i \text{ odd}, \\ -(L/2 - 2\log 2) + \varepsilon_i, & i \text{ even}, \end{cases}$$

and $L/2 - 2\log 2 \ge \log 2 > 4/63 \ge |\varepsilon_i|$ for $R \ge 64$. As $u \to 0^+$, $f(u) = \log u + O(1) \to -\infty$. As $u \to \infty$, $f(u) = (1 + \sum_{m=1}^{n} 2(-1)^m)\log u + o(\log u)$ with $1 + \sum_{m=1}^{n} 2(-1)^m = -1$ (n odd) or $+1$ (n even). So $f - c$ takes alternating signs at $n+2$ points $0^+ < R < R^2 < \cdots < R^n < \infty$, has at least $n+1$ zeros, hence exactly $n+1$, all simple, by Lemma 6. □

Proof (of Theorem 1)**.** Take $\mathcal{N}_n(\alpha,\beta)$ with $\alpha_j = 64^{j-1}$, $\beta_j = 1$, and rate constants $\kappa_j = 1$ for $j \neq 1$ and $\kappa_1 = e^c$ with c as in Proposition 7 ($w_1 = 1$, so $\sum_j w_j \log \kappa_j = c$). By Lemmas 4 and 5 and Proposition 7 the network has exactly $n+1$ positive equilibria, all nondegenerate. By Lemma 3 it is a full-rank quadratic $(n, n+2, n)$ network with $n+2$ distinct sources, dynamically nontrivial and not source deficient. In fact κ_1 is rational: $\sum_{m=1}^{n}(-1)^m m$ equals $-(n+1)/2$ for odd n and $n/2$ for even n, so $M_1 = -nL$ or $(n+1)L$, and $\kappa_1 = e^c = 64^{\mp(n+1/2)} = 2^{\mp(6n+3)}$ (upper sign for odd n). □

4 Small explicit examples

The separation $R = 64$ is convenient for a uniform proof but far from necessary. For $n = 3$, the network $\mathcal{N}_3(\alpha,\beta)$ with $(\alpha_2,\alpha_3,\alpha_4) = (1,3,16)$ and $(\beta_2,\beta_3,\beta_4) = (14,3,1)$,

$$0 \to X_1{+}3X_2{+}16X_3,\ \ 2X_1 \to 16X_1{+}3X_2{+}X_3,\ \ X_1{+}X_2 \to X_2,\ \ X_2{+}X_3 \to X_3,\ \ X_3 \to 0,$$

whose products have molecularity at most 20, has four positive nondegenerate equilibria for $\kappa_1 = 1/1600$ (all other rate constants equal to 1); the certificate is described below. A quadratic $(3,5,3)$ network outside this family, found by a computer search, is $Z \to 78Y$, $2Z \to Z$, $2Y \to X+Y$, $X \to 0$, $X+Y \to 24X+20Z$; it also has four positive nondegenerate equilibria, for instance with rate constant $9971888893/778$ for $2Z \to Z$ and 1 for the other reactions (`scripts/other_example_n3.py`). Neither example has minimal product molecularity: the $(3,5,3)$ example of [2], $X \to 0$, $Y \to 6X+4Z$, $2X \to 6Y+4Z$, $2Z \to 4Y$, $Y+Z \to 0$ with rate constant $2/3137$ for $2Z \to 4Y$ and 1 otherwise, has products of molecularity at most 10 and four positive nondegenerate equilibria (checked in `verification/prior_art_check.py`).

5 Prior work

The question of Remark 5.8 had already been answered when this note was written, and the algebraic core of the construction is older.

- *Circuits.* A system of n real polynomial equations in n variables whose supports lie in a set of $n+2$ monomials (a circuit) has at most $n+1$ nondegenerate positive solutions, and this bound is attained [5]; this is the polynomial counterpart of Theorem 5.5 of [3]. The reduction to one variable used in Lemmas 4–6 is the standard Gale-duality argument for circuits.

- *The same polynomial system.* Phillipson and Rojas Theorem 1.6 [1] prove that the system G_ε: $x_1x_2 = \varepsilon + x_1^2$, $x_ix_{i+1} = 1 + \varepsilon^{2i-3}x_1^2$ $(2 \le i \le n-1)$, $x_n = 1 + \varepsilon^{2n-3}x_1^2$ has exactly $n+1$ positive roots, all nondegenerate, for $\varepsilon = 1/4$. After dividing the i-th equilibrium equation by κ_{i+1}, this is precisely the equilibrium system of $\mathcal{N}_n(\alpha,\beta)$ with $\alpha_j = 16^{j-2}$, $\beta_j = 1$ and rate constants $\kappa_0 = 1/4$, $\kappa_1 = \kappa_2 = 1$, $\kappa_{i+1} = 4^{2i-3}$ $(2 \le i \le n)$, i.e. geometric ratio $R = 16$ instead of 64 (checked exactly for $n = 2,\dots,6$ in `verification/prior_art_check.py`). In the same way, $\mathcal{N}_n$ with $\alpha_j = 64^{j-1}$, $\beta_j = 1$, $\kappa_0 = 1/512$, $\kappa_1 = \kappa_2 = 1$, $\kappa_{i+1} = 8^{2i-3}$ $(2 \le i \le n)$ has equilibrium equations $G_{1/8}$, and these rate constants give the same constant $c = \sum_j w_j \log \kappa_j = \mp(6n+3)\log 2$ as those used in the proof of Theorem 1. So the $R = 64$ networks of Theorem 1 are, through the bijection of Lemma 4, the Phillipson–Rojas systems with $\varepsilon = 1/8$, and Theorem 1 is essentially Theorem 1.6 [1] plus the observation that $\mathcal{N}_n(\alpha,\beta)$ realizes G_ε as a mass-action network. Section 3 gives a short self-contained proof over $\mathbb{R}$.

- *The chemical answer.* An anonymous, unrefereed manuscript posted on Zenodo and GitHub on 4 October 2026 [2] realizes the Phillipson–Rojas chain as a quadratic $(n, n+2, n)$ mass-action network for every $n \geq 2$ (with sources $0, X_1, X_i + X_{i+1}, 2X_n$, the same support as $\mathcal{N}_n$ after reversing the order of the species) and explicitly answers Remark 5.8; it also gives the $(3, 5, 3)$ example with product molecularity 10 quoted above. Its explicit networks for $n = 2, \ldots, 6$ were re-checked exactly for this note (`verification/prior_art_check.py`).

The construction and proof in Sections 1–3 do not rely on these works, which were not cited in the first version of this note and were identified during its independent verification.

6 Computer verification

The proof above needs no computer. As a check, `scripts/certify_family.py` builds $\mathcal{N}_n(\alpha, \beta)$ for the small $n = 3$ example, for $0 \to X$, $2X \to 3X$, $X \to 0$, and for the $R = 64$ family with $n = 1, \ldots, 8$, and verifies in exact arithmetic: quadratic distinct sources, nonnegative products, $\operatorname{rank} \Gamma = n$, $\operatorname{rank}[S|\mathbf{1}] = n + 1$, $\Gamma\alpha = \Gamma\beta = 0$, the dependency w, that f' has exactly n positive zeros, and that for an explicit rational C the polynomial $\prod_{w_j>0} v_j^{w_j} - C \prod_{w_j<0} v_j^{-w_j}$ has exactly $n + 1$ positive roots, all simple (Sturm sequences and $\gcd(P, P') = 1$); it also computes the corresponding equilibria to 80 digits and checks $\Gamma\rho(x) = 0$ and $\det J(x) \neq 0$ (100 checks, all pass).

Independent verification. The scripts in `verification/` do not use the reduction to one variable. `direct_check.py` writes each network as a list of reactions, solves the mass-action equations with exact rational rate constants by msolve (exact Gröbner basis and certified real root isolation), and shows with Singular that the ideal generated by the equations, $\det Dg$ and $t\, x_1 \cdots x_n - 1$ is the unit ideal, so that no equilibrium with nonzero coordinates is degenerate; a Sturm count (PARI/GP) on the eliminant in x_1 cross-checks the number of positive equilibria. It confirms $n + 1$ positive nondegenerate equilibria for the $R = 64$ family with exactly the rate constants of the proof ($\kappa_1 = 2^{\mp(6n+3)}$), $n = 1, \ldots, 10$, and four for the two $(3, 5, 3)$ examples of this note ($\mathcal{N}_3$ with $\kappa_1 = 1/1600$, and the computer-found network). `prior_art_check.py` applies the same pipeline to the networks of [2] and to the realizations of $G_{1/4}$ and $G_{1/8}$, and `uniform_signs_arb.py` checks the sign pattern of Proposition 7 in ball arithmetic for $n \leq 300$.

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